

DATASET DESCRIPTION

Dataset Name: VinBigData Chest X-ray Abnormalities Detection

Overview: The dataset consists of chest X-ray images collected for the purpose of detecting thoracic abnormalities. It is used for training deep learning models for abnormality detection and classification. Early detection of chest abnormalities can assist radiologists in faster and more accurate diagnosis. In this project, the dataset is used to develop an AI-powered system that can automatically detect and localize lung abnormalities from chest X-ray images using deep learning techniques.

Source: The dataset was obtained from Kaggle and accessed through a Kaggle notebook.

Domain: HealthCare- Medical Imaging, Computer Vision

Number of samples: 18000 CXR scans covering 14 critical radiographic findings.

Data Distribution: The dataset includes 14 abnormality classes with varying numbers of samples per category. This results in potential class imbalance, which may influence model training and performance. Some abnormalities are more frequently represented than others.

Column Annotations:

- Columns: image_id - unique image identifier
- class_name - the name of the class of detected object
- class_id - the ID of the class of detected object
- rad_id - the ID of the radiologist that made the observation
- x_min , y_min – Minimum coordinates of object bounding box
- x_max , y_max - Maximum coordinates of object bounding box

Target variable: Class_name which specifies the name of class of the detected object.

File Format:

Images: DICOM

Annotations: CSV

Data Preprocessing: DICOM images were converted to PNG/JPEG format, converted to greyscale images, resized to maintain uniform dimensions, and normalized to standardize pixel intensity values.

